

TSURUGA, Japan

WMO: 476310

Lat: 35.6533N

Lon: 136.0622E

Elev: 12

StdP: 101.18

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	0.3	-6.3	2.2	4.6	-5.1	2.5	4.4	13.3	3.8	12.3	3.6	3.7	180	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.7	33.8	25.1	32.7	25.1	31.6	24.8	26.3	31.2	25.7	30.6	25.3	30.0	5.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.1	29.2	24.4	19.4	28.8	23.9	18.8	28.4	82.1	31.3	79.7	30.7	77.6	30.1	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.9	8.8	-2.3	35.7	1.1	0.8	-3.0	36.3	-3.7	36.8	-4.3	37.2	-5.1	37.8		
(5)			-3.4	27.3	1.1	0.7	-4.2	27.8	-4.8	28.2	-5.5	28.6	-6.3	29.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	5.0	8.2	13.4	18.3	22.2	26.7	28.2	24.0	18.4	12.6	7.4
	DBStd	8.54	2.31	2.70	3.27	3.45	2.90	2.51	2.63	2.18	2.90	2.79	3.03	2.99
	HDD10.0	496	168	142	77	9	0	0	0	0	0	0	9	92
	HDD18.3	1851	426	372	315	153	37	1	0	0	1	33	173	341
	CDD10.0	2614	1	3	20	110	258	366	517	563	421	259	87	10
	CDD18.3	927	0	0	0	3	36	117	258	305	172	34	1	0
	CDH23.3	8074	0	0	0	18	158	580	2526	3448	1244	101	0	0
	CDH26.7	2807	0	0	0	2	20	113	914	1423	329	6	0	0
Wind	WSAvg	4.1	4.5	4.7	4.6	4.4	4.2	3.7	3.6	3.7	3.6	3.7	4.1	4.4
Precipitation	PrecAvg	2093	237	152	137	114	148	166	224	146	199	142	162	255
	PrecMax	2769	431	228	225	189	456	303	455	388	418	287	324	547
	PrecMin	1487	77	46	69	43	62	72	69	29	72	59	87	125
	PrecStd	275	90	44	42	43	88	59	99	82	96	48	49	99
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	15.6	20.2	25.8	28.9	31.2	34.9	35.4	33.0	27.6	22.0	17.1
		MCWB	9.3	11.5	12.6	15.3	18.8	22.9	25.1	25.1	24.3	21.3	15.9	13.0
	2%	DB	11.1	13.0	17.6	22.7	26.7	29.4	33.6	34.1	31.5	25.7	20.3	15.1
		MCWB	7.7	9.1	11.5	14.8	17.8	22.1	25.1	25.2	24.0	20.3	15.4	11.0
	5%	DB	9.8	11.4	15.5	20.9	25.1	27.9	32.4	33.2	29.9	24.3	18.9	13.6
		MCWB	6.5	7.8	10.6	14.5	17.3	21.5	25.0	25.3	23.6	19.1	14.5	10.0
	10%	DB	8.6	9.8	13.8	19.1	23.4	26.5	31.1	32.1	28.5	23.0	17.7	12.2
		MCWB	5.5	6.4	9.6	13.6	16.9	21.0	24.8	25.2	23.0	18.1	13.9	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	12.2	14.3	18.9	21.2	24.8	27.2	27.1	25.7	22.9	17.7	14.2
		MCDB	12.1	14.9	17.4	21.8	25.1	28.8	31.2	32.5	30.3	26.4	20.1	16.1
	2%	WB	8.2	9.8	12.8	16.6	19.6	23.7	26.2	26.5	25.0	21.2	16.5	11.8
		MCDB	10.3	12.3	16.3	20.5	23.8	27.4	30.9	31.7	29.3	24.6	19.0	14.2
	5%	WB	7.0	8.3	11.4	15.3	18.6	22.8	25.7	26.0	24.3	20.1	15.4	10.5
		MCDB	9.4	10.5	14.6	19.3	22.8	26.4	30.4	31.1	28.5	22.9	18.1	13.1
	10%	WB	5.8	6.8	10.1	14.2	17.8	21.9	25.1	25.6	23.7	19.0	14.3	9.2
		MCDB	8.2	9.5	13.4	17.8	22.1	25.3	29.9	30.4	27.7	22.1	17.3	11.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.9	7.3	7.9	7.7	6.3	6.1	6.7	6.6	7.1	6.8	5.8
		MCDBR	7.1	8.3	10.1	10.3	10.5	8.5	7.9	7.7	7.6	8.0	7.9	7.5
		MCWBR	4.9	5.9	6.5	5.3	4.8	3.8	3.0	2.9	3.2	4.0	4.9	5.1
	5% WB	MCDBR	6.3	7.0	8.1	7.9	7.8	6.4	7.1	7.0	6.6	6.1	6.3	6.6
		MCWBR	5.0	5.8	6.2	4.8	4.5	3.3	3.2	3.1	3.2	3.8	4.8	5.2
Clear-Sky Solar Irradiance	taub	0.437	0.469	0.534	0.560	0.564	0.587	0.581	0.552	0.498	0.461	0.438	0.434	
	taud	1.996	1.938	1.819	1.813	1.863	1.880	1.952	2.024	2.136	2.140	2.101	2.027	
	Ebn at Noon	718	747	735	743	747	728	729	742	763	751	716	690	
	Edh at Noon	137	163	199	211	203	200	185	169	144	131	123	125	
All-Sky Solar Radiation	RadAvg	1.86	2.57	3.66	4.63	5.17	4.81	4.82	5.00	3.93	3.17	2.24	1.67	
	RadStd	0.14	0.25	0.30	0.41	0.55	0.41	0.73	0.53	0.34	0.28	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+44
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page